

Chetan Goenka

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EDUCATION

University of California, Berkeley

Berkeley, CA

Master of Science in Electrical Engineering & Computer Science

Expected May 2026

Bachelor of Arts in Computer Science, GPA: 3.90/4.0, Graduated with Honors

Dec 2024

Relevant Coursework: Efficient Algorithms, Data Structures, Artificial Intelligence, Deep Neural Networks, User Interface Development, Internet Architecture and Protocol, Computer Architecture, Machine Learning & Human Behavior

SKILLS

Programming Languages: Python, Java, SQL, TypeScript, JavaScript, C, C++, C#, HTML, CSS, Swift, R, Bash, Assembly

Frameworks/Tools: Pandas, NumPy, Git, React, Next.js, Tailwind, OpenAI/Claude, AWS, PostgreSQL, Figma, Pytest, Docker

RELEVANT EXPERIENCE AND PROJECTS

UC BERKELEY EECS DEPARTMENT - HCI Research Lab

Berkeley, CA

Engineering Design Scholar & Research Assistant

June 2024 – Present

- Engineered a web-based coding environment using React, CodeMirror, and OpenAI API, delivering dynamic, context-aware code explanations (line-by-line, block-wise, and data-flow) to enhance developer comprehension of AI-generated code
- Designed and implemented structured JSON metadata extraction with Zod, ensuring accurate line mappings, logical groupings, and data-flow tracking, enabling dynamically linked LLM-generated explanations and interactive overlays
- Extended CodeMirror's annotation capabilities by building custom DOM-based overlays for code elements, hover-triggered tooltips, and click-driven interactions, improving code navigation and readability through dynamic in-editor explanations
- Conducted pilot study with developers to evaluate tool effectiveness, analyzing user interaction data and feedback to validate design decisions and measure comprehension improvements

PYTHON COMPILER

Jan 2024 – May 2024

- Built a compiler using Java for a statically typed dialect of Python, implementing key stages including lexical analysis, syntax analysis, and optimized code generation for RISC-V assembly
- Constructed a code generator to translate type-annotated ASTs into efficient RISC-V assembly code, integrating techniques such as constant propagation, dead code elimination, and register allocation, reducing runtime by 40%
- Implemented list operations, function calls, and control flow structures, ensuring correct execution of python programs on a RISC-V simulator, and constructed an integrated semantic analyzer and type checker to maintain well-typed programs

UC BERKELEY EECS DEPARTMENT - Programming Course (Data 100)

Berkeley, CA

Undergraduate Course Staff

Aug 2023 – May 2024

- Provided weekly programming guidance to 25+ students in Python and data manipulation libraries, helping debug complex programming projects and resolve syntax errors, logic bugs, and algorithmic challenges
- Created comprehensive troubleshooting strategies and technical study materials for common programming challenges, facilitating collaborative problem-solving in office hours, online forums, and discussion sections
- Evaluated 5000+ student assignments using structured rubrics and provided detailed code feedback, ensuring adherence to coding standards and technical requirements through systematic review processes across multiple semesters

ORACLE CORPORATION

Austin, TX

Data Science Intern

Jun 2022 – Aug 2022

- Developed a customer scoring system using Python, pandas, and scikit-learn with regression and AdaBoost algorithms, collaborating with team members to increase marketing efficiency by 25% across enterprise accounts
- Enhanced data preprocessing and model deployment pipeline for production systems through data cleaning and feature engineering techniques, collaborating with senior engineers to decrease overall run time by 18%
- Analyzed marketing campaign data from hundreds of customer interactions using Python and SQL to extract key insights, creating visualizations and reports to communicate findings to sales and marketing teams